

Comparison across models

Methods

For each method, we evaluated 1-step and 2-step ahead forecasts, using the last 5 years of the dataset (2015 - 2019) as test data (e.g. for the 1-step ahead forecast, years up to 2014 were used to predict 2015, years up to 2016 were used to predict 2017, etc.). Total RMSE was calculated for each test year ($n = 5$), and RMSE was then averaged across test years for a final RMSE statistic. Finally, we also ran the models with 0-step ahead forecasts and evaluated the RMSE on the last year of data (e.g. data through 2016 used, and RMSE calculated for 2016); this latter approach is useful in evaluating how forecast skill degrades for out of sample versus in sample data.

Results

- RF seems to be doing better than GAM
- GAMs do as well with forecasts as training data (perhaps because not very well overall)
- Delta - RF approach with separate models for classification and prediction seem to do worse than a combined RF model (this is kind of surprising)

method	years.ahead	rmse
random forest	0	152.639
random forest	1	350.204
random forest	2	349.980

method	years.ahead	rmse
gam	0	410.433
gam	1	415.036
gam	2	415.592

method	years.ahead	rmse
random forest - delta	0	200.140
random forest - delta	1	359.746
random forest - delta	2	366.884